Question 1 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	3^6	1	B1
(b)	5^3	1	B1
(c)	2^{12}	1	B1

Question 2 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	1	1	B1
(b)	$\frac{1}{16}$	1	B1
(c)	$\frac{3}{8}$	1	B1

Question 3 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	5	1	B1
(b)	3	1	B1
(c)	8	2	B2 for correct answer, or M1 for $(\sqrt[4]{16})^3$ or 2^3 or $\sqrt[4]{4096}$

Question 4 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	2^2 or 4	1	B1
(b)	6^3 or 216	1	B1
(c)	$2\frac{1}{4}$	2	M1 for $\left(\frac{3}{2}\right)^2$ or $\frac{1}{\left(\frac{2}{3}\right)^2}$ or $\frac{2^{-2}}{3^{-2}}$ or $\frac{9}{4}$

Question 5 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	$\frac{1}{2}$	2	B2 for correct answer, or M1 for $\frac{1}{8^{\frac{1}{3}}}$ or $\frac{1}{2}$ seen in working
(b)	$\frac{2}{3}$	2	B2 for correct answer, or M1 for $\left(\frac{4}{9}\right)^{\frac{1}{2}}$ or $\frac{1}{\sqrt{\frac{9}{4}}}$ or $\frac{3}{2}$ seen

Question 6 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	4	2	M1 for $8^{\frac{2}{3}}$ or $(\sqrt[3]{8})^2$ or 2^2 or $\left(\frac{1}{4}\right)^{-1}$
(b)	$\frac{5}{8}$	2	M1 for $125^{\frac{1}{3}} = 5$ or $2^{-3} = \frac{1}{8}$ or $\frac{5}{2^3}$

Question 7 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	-4	2	M1 for $3^x = 3^{-4}$ or $\frac{1}{81} = \frac{1}{3^4}$
(b)	$\frac{7}{4}$	2	M1 for $(2^2)^{2x-1} = 2^5$ or $2(2x-1) = 5$ or $4x - 2 = 5$

Question 8 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	12	3	M2 for $\frac{2^5 \times 2^3 \times 3^3}{3^2 \times 2^6}$ or M1 for breaking down bases into prime factors: $6^3 = 2^3 \times 3^3$ or $4^3 = 2^6$
(b)	1	2	M1 for $\left(5^{\frac{1}{2}}\right)^4 = 5^2$ and $10^2 = 2^2 \times 5^2$ leading to $\frac{5^2 \times 2^2 \times 5^2}{2^2 \times 5^3}$ or $\frac{25 \times 100}{4 \times 125}$

Question 9 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	$\frac{27}{8}$	2	M1 for $\left(\frac{81}{16}\right)^{\frac{3}{4}}$ or $\left(\frac{3}{2}\right)^3$ or $\frac{2^{-3}}{3^{-3}}$
(b)	$\frac{1}{17}$	3	M1 for $4^{\frac{3}{2}} = 8$, M1 for $27^{\frac{2}{3}} = 9$, then leading to $(8+9)^{-1}$

Question 10 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	$\frac{3}{5}$	2	M1 for converting to improper fraction $\left(\frac{25}{9}\right)^{-\frac{1}{2}}$ or finding $\sqrt{\frac{9}{25}}$
(b)	4	2	M1 for writing all terms with base 3: $\frac{3^{-5}\times(3^2)^2}{(3^3)^{-1}}$ or $3^{-5+4-(-3)}$ or 3^2

Question 11 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
x, y	$x=1, y=-2$	4	M1 for $x+2y=-3$, M1 for $3x+y=1$, M1 for attempting to solve the simultaneous linear system, A1 for both values correct.

Question 12 (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
(a)	3×4^{-x}	2	M1 for $243^{\frac{x}{5}}=(3^5)^{\frac{x}{5}}=3^x$ leading to $\frac{3^x\times 4^{-x}}{3^{x-1}}$ or $3^{x-(x-1)}\times 4^{-x}$
(b)	0.0712	2	M1 for $3\times 4^{-2.7}$ or $3\times 0.0237...$ or $0.07119...$ seen